
CONTRASTIVE REPRESENTATION LEARNING FOR PROMPT-RESPONSE ALIGNMENT

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ABSTRACT

Reinforcement learning from human feedback (RLHF) has become a standard approach for aligning large language models with human preferences, but reward models may rely on superficial features such as length, style, or fluency rather than true adherence to prompt intention. This thesis studies prompt-response alignment as a representation learning problem and proposes TextCLIP, a contrastive framework that learns a shared embedding space for prompts and responses. Instead of measuring absolute response quality, TextCLIP learns a prompt-conditioned compatibility function that captures how well a response addresses the semantic goal and constraints of a given prompt.

TextCLIP uses dual encoders with projection heads and is trained with a symmetric contrastive objective that pulls matched prompt-response pairs together while pushing mismatched pairs apart. I evaluate both a lightweight TextCLIP model trained from scratch and a pretrained variant built on BAAI/bge-base-en-v1.5. On the hardest corresponding-rewrite task, the original TextCLIP achieves 0.768 accuracy, while the pretrained TextCLIP reaches 0.856, outperforming a pretrained embedding similarity baseline. Length-bias analysis further suggests that the model does not simply reward longer responses when added content becomes off-topic.

I also apply DPO-style fine-tuning to adapt the shared embedding space to on-topic, toxicity-related, and general preference tasks. Fine-tuning substantially improves pairwise accuracy and makes the separation between chosen and rejected responses more pronounced in UMAP visualizations. Overall, these results suggest that contrastive representation learning provides a useful and interpretable framework for modeling prompt-specific alignment.

1 INTRODUCTION

Reinforcement Learning from Human Feedback (RLHF) has become a widely adopted framework for aligning large language models (LLMs) with human preferences (Christiano et al., 2017a). But it also introduces the risk of reward hacking. Instead of truly following user intent, models may learn to exploit weaknesses in the reward model to obtain higher scores. In particular, reward models often rely on superficial features such as response length, fluency, or style, leading to biases like length bias (Amodi et al., 2016). As a result, a response can receive a high reward even if it does not meaningfully address the prompt.

Prior work often tries to remove these specific spurious artifacts. However, features such as length are not always irrelevant. In many cases, they are part of the prompt intention itself—for example, when a user asks for a detailed explanation or a brief answer. This means the key problem is not simply to eliminate particular artifacts, but to measure how well a response addresses the intention of the prompt. More broadly, we need a function that captures whether a response is truly on-topic for the given prompt.

This thesis studies that question through a contrastive representation learning framework, referred to as TextCLIP, for modeling prompt-response alignment. Inspired by CLIP, which learns a shared embedding space for paired images and texts, TextCLIP aims to learn a joint embedding space in which matched prompts and responses are placed close together, while mismatched pairs are pushed

apart. The underlying intuition is that contrastive learning can encourage the model to capture relational information: a good response is not defined in isolation, but by its compatibility with a particular prompt. By training on prompt–response pairings rather than only scalar preference labels, the model may be able to learn a representation that better reflects prompt-conditional semantic alignment.

The motivation for this approach is twofold. First, contrastive learning provides a natural way to model prompt-specific compatibility. For each prompt, matched responses can be treated as positive examples, while responses to other prompts serve as negatives. This shifts the learning problem away from fitting broad preference patterns and toward identifying which responses actually reflect the intention of a given prompt. Under this setup, surface features are not removed by assumption; instead, the model is encouraged to distinguish features that are genuinely related to the prompt from those that are merely incidental. Second, a shared embedding space offers an explicit representation of prompt–response alignment. Such a representation is useful not only for scoring, but also for retrieval, response ranking, and analyzing how alignment signals are organized geometrically. If the learned space captures prompt intention and response relevance, it can also help reveal when standard reward models rely on factors such as length or style for the wrong reasons.

The central hypothesis of this thesis is that contrastive learning can provide a more robust signal of prompt–response alignment than objectives that depend primarily on direct preference supervision. Specifically, I investigate whether a CLIP-style architecture can learn representations that distinguish semantically aligned responses from misaligned ones, while being less sensitive to superficial artifacts. This thesis therefore positions prompt–response alignment as a representation learning problem: the goal is not only to score outputs, but to construct an embedding space in which semantic compatibility between prompts and responses becomes geometrically meaningful.

To study this problem, I develop a TextCLIP framework in which prompts and responses are encoded separately and projected into a shared latent space. The model is trained using a contrastive objective that encourages matched prompt–response pairs to have similar representations. In settings where multiple responses are available for the same prompt, these responses provide a particularly useful training signal, as they allow the model to capture prompt-level semantic consistency while contrasting against responses generated for different prompts. Through this framework, I examine whether the learned representations can serve as an alternative basis for measuring “on-topicness,” and whether they reduce the influence of spurious features that commonly affect preference-based evaluation.

This thesis makes three main contributions. First, it formulates prompt–response alignment as a contrastive representation learning problem and proposes TextCLIP as a CLIP-inspired framework for learning a shared embedding space for prompts and responses. Second, it investigates the extent to which such a framework can capture semantic alignment while mitigating reliance on confounding features such as response length or stylistic variation. Third, it provides an empirical study of the behavior of the learned representations, with particular attention to whether contrastive training offers a useful signal for evaluating how well a response matches the intention of a given prompt.

2 PRELIMINARY

2.1 REINFORCEMENT LEARNING FROM HUMAN FEEDBACK

Reinforcement Learning from Human Feedback (RLHF) (Christiano et al., 2017b; Ziegler et al., 2019) has become a key technique for integrating human preference signals into machine learning methods. In particular, RLHF has become a standard component in the post-training pipeline of foundation Large Language Models (LLMs), which serves to align the outputs of these models with human values such as helpfulness, harmlessness, and honesty (Ouyang et al., 2022; Bai et al., 2022). Notable examples include the revolutionary closed-source ChatGPT (Achiam et al., 2023), Claude (Anthropic, 2023), and Gemini (Team et al., 2023), as well as the powerful open-source models like Zephyr (Tunstall et al., 2023) and LLaMA-3 (Meta, 2024). In particular, since the introduction of ChatGPT, RLHF has attracted significant interest in a diverse set of communities.

However, recent work has revealed that reward models are susceptible to reward hacking, where models exploit imperfections in the learned reward function rather than genuinely aligning with

human intent (Amodei et al., 2016). Reward hacking can arise from unintentional and prompt-irrelevant human preferences (Wang et al., 2025).

Preference Data and Reward Modeling Let $x \in \mathcal{X}$ denote a prompt and $y \in \mathcal{Y}$ denote a response. A preference dataset is typically represented as

$$\mathcal{D}_{\text{pref}} = \{(x_i, y_i^w, y_i^l)\}_{i=1}^n,$$

where y_i^w is the preferred or chosen response and y_i^l is the less preferred or rejected response for the same prompt x_i . The preference relation can be written as

$$y_i^w \succ y_i^l \mid x_i,$$

meaning that, conditional on prompt x_i , the response y_i^w is judged to be better aligned than y_i^l .

This pairwise formulation is useful because it avoids requiring annotators to assign an absolute score to each response. Instead, annotators only need to compare two candidate responses under the same prompt. The resulting supervision is inherently prompt-conditional: a response is not judged in isolation, but relative to another response for the same user request.

A common objective is the Bradley–Terry style loss (Bradley and Terry, 1952):

$$\mathcal{L}_{\text{RM}}(\phi) = -\mathbb{E}_{(x, y_w, y_l)} [\log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l))], \quad (1)$$

where $\sigma(\cdot)$ is the logistic sigmoid function.

Direct Preference Optimization Direct Preference Optimization (DPO) is an alternative to the traditional two-stage RLHF pipeline. Instead of first training a reward model and then using reinforcement learning to optimize a language model policy, DPO directly optimizes the policy on preference pairs (Zhao et al., 2023).

Let $\pi_\theta(y \mid x)$ denote the language model policy being optimized, and let $\pi_{\text{ref}}(y \mid x)$ denote a fixed reference policy, often initialized from a supervised fine-tuned model. For each preference triple (x, y^w, y^l) , DPO encourages the policy to increase the relative likelihood of the chosen response y^w over the rejected response y^l , while controlling deviation from the reference policy.

The DPO objective can be written as

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x, y^w, y^l) \sim \mathcal{D}_{\text{pref}}} \left[\log \sigma \left(\beta \left[\log \frac{\pi_\theta(y^w \mid x)}{\pi_{\text{ref}}(y^w \mid x)} - \log \frac{\pi_\theta(y^l \mid x)}{\pi_{\text{ref}}(y^l \mid x)} \right] \right) \right],$$

where $\beta > 0$ controls the strength of the preference optimization. Intuitively, this loss encourages the current policy to improve the relative probability of the preferred response compared with the reference model, while reducing the relative probability of the rejected response.

In standard DPO, the optimized object is an autoregressive language model policy $\pi_\theta(y \mid x)$. In this thesis, we do not use TextCLIP similarity as a reward to optimize a language model policy. Instead, we apply a DPO-style preference tuning idea to the TextCLIP encoders themselves. The goal is to shape the shared embedding space so that preferred responses are closer to their corresponding prompts than rejected responses.

2.2 CONTRASTIVE LEARNING

Contrastive learning is an instance-based (Wu et al., 2018) unsupervised learning method (Jaiswal et al., 2020). Contrastive learning learns common features between similar instances and distinguishes different features among dissimilar instances. In feature space, contrastive learning brings similar instances closer together and pushes dissimilar instances farther apart. Rather than directly predicting a scalar label, the goal is to construct an embedding space in which related examples are close and unrelated examples are far apart.

General setup. Let u and v denote two views of related information. In the standard contrastive setting, an encoder maps them to vector representations

$$z_u = f(u), \quad z_v = g(v), \quad (2)$$

where $f(\cdot)$ and $g(\cdot)$ may be identical or separate encoders. Their similarity is often measured by cosine similarity,

$$\text{sim}(z_u, z_v) = \frac{z_u^\top z_v}{\|z_u\| \|z_v\|}. \quad (3)$$

For an anchor example u_i , let v_i^+ denote its positive pair and let $\{v_j^-\}_{j \neq i}$ denote negative examples. A standard InfoNCE objective is

$$\mathcal{L}_{\text{NCE}} = - \sum_{i=1}^N \log \frac{\exp(\text{sim}(z_{u_i}, z_{v_i^+})/\tau)}{\sum_{j=1}^N \exp(\text{sim}(z_{u_i}, z_{v_j})/\tau)}, \quad (4)$$

where $\tau > 0$ is a temperature parameter.

This loss increases the similarity of positive pairs while reducing the relative similarity of negative pairs. The resulting representation space is shaped by pairwise correspondence rather than by direct supervision on handcrafted features.

Alignment and uniformity. A useful perspective on contrastive learning is that it balances two principles: *alignment*, which pulls positive pairs together, and *uniformity*, which spreads representations across the space. Alignment encourages semantically related examples to cluster, while uniformity prevents degenerate collapse in which all representations become identical. This makes contrastive learning well suited for settings where the key object of interest is relational structure.

CLIP. Contrastive Language–Image Pre-training (CLIP) (Radford et al., 2021) learns aligned representations across modalities via a contrastive objective. Using a dual-encoder architecture, CLIP maps images and texts into a shared embedding space, where matched pairs are encouraged to have high similarity and mismatched pairs are pushed apart. This approach provides a general framework for learning semantic alignment through contrastive representation learning.

2.3 CONTRASTIVE LEARNING FOR PROMPT–RESPONSE ALIGNMENT

Contrastive representation learning has been theoretically characterized by two key principles: *alignment*, which encourages representations of positive pairs to be close, and *uniformity*, which promotes a well-distributed representation space (Wang and Isola, 2020). This perspective provides a useful foundation for our setting, where prompt–response alignment can be viewed as learning representations that preserve semantic correspondence while remaining robust to spurious correlations.

The proposed framework is based on the idea that prompt–response alignment can be formulated as a contrastive correspondence problem. Given a prompt p_i and a set of responses $\{r_{ij}\}_{j=1}^K$, matched prompt–response pairs are treated as positive examples, while responses associated with other prompts are treated as negatives.

Let

$$h_i = f(p_i), \quad g_{ij} = g(r_{ij}) \quad (5)$$

denote the prompt and response embeddings. Their similarity is defined as

$$s(p_i, r_{kj}) = \frac{h_i^\top g_{kj}}{\tau}. \quad (6)$$

For each prompt p_i , responses $\{r_{ij}\}_{j=1}^K$ generated for that same prompt are treated as positives, while responses from other prompts form the negative pool. A natural objective is therefore

$$\mathcal{L} = - \frac{1}{NK} \sum_{i=1}^N \sum_{j=1}^K \log \frac{\exp(s(p_i, r_{ij}))}{\sum_{k=1}^N \sum_{m=1}^K \exp(s(p_i, r_{km}))}. \quad (7)$$

Unlike reward modeling, this objective does not ask the model to produce an absolute scalar judgment of response quality. Instead, it learns a representation in which a response is evaluated relative to the prompt it is meant to answer. This is especially useful in our setting, because features such as length or style need not be treated as universally undesirable. Their relevance depends on whether they help express the intention of the prompt.

3 METHODOLOGY

3.1 PROBLEM SETUP

Let

$$\mathcal{D} = \{(p_i, \mathcal{R}_i)\}_{i=1}^N$$

denote a dataset of prompts and responses, where p_i is the i -th prompt and

$$\mathcal{R}_i = \{r_{i1}, r_{i2}, \dots, r_{iK}\}$$

is the set of K responses associated with prompt p_i . In our setting, these responses are generated by different large language models for the same prompt.

The goal of TextCLIP is to learn a similarity function $s(p, r)$ that measures the extent to which a response r addresses the underlying intention of a prompt p . Rather than modeling absolute response quality in isolation, this function is designed to capture prompt-conditioned alignment. Moreover, the learned similarity goes beyond surface-level semantic similarity, such as keyword matching or broad topical relatedness, and instead seeks to encode whether the response satisfies the specific communicative goal, constraints, and intent expressed by the prompt.

This distinction is important in our setting. Features such as response length, style, or formatting are not always spurious. Their relevance depends on the prompt itself. A long response may be undesirable for one prompt but appropriate for another if the user requests a detailed explanation. Therefore, rather than explicitly removing such features, we seek to learn a representation in which compatibility is determined by whether a response reflects the meaning and intention of the prompt.

3.2 OVERVIEW OF TEXTCLIP

TextCLIP is inspired by CLIP, which learns a shared embedding space for image–text pairs. Here, we adapt the same idea to prompt–response pairs. The framework is shown in Fig 1, consisting of two encoders:

- a **prompt encoder** $f_\theta(\cdot)$,
- a **response encoder** $g_\phi(\cdot)$.

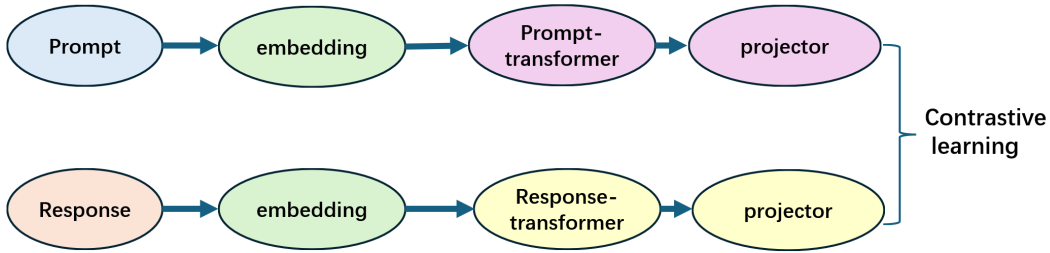


Figure 1: The framework of TextCLIP.

Given a prompt p and a response r , the two encoders map them into a common d -dimensional embedding space:

$$\mathbf{h}_p = f_\theta(p) \in \mathbb{R}^d, \quad \mathbf{h}_r = g_\phi(r) \in \mathbb{R}^d.$$

The similarity between the prompt and the response is then defined in this shared space. During training, matched prompt–response pairs are encouraged to have high similarity, while mismatched pairs are pushed apart.

3.3 ENCODER ARCHITECTURE

Input Representation Each prompt and response is first tokenized into a sequence of tokens. Let

$$p = (w_1^{(p)}, w_2^{(p)}, \dots, w_{T_p}^{(p)})$$

and

$$r = (w_1^{(r)}, w_2^{(r)}, \dots, w_{T_r}^{(r)})$$

denote the tokenized prompt and response, respectively.

Each token is mapped to an embedding vector, and positional embeddings are added to preserve word order. For both prompt and response, the token sequence is then passed through a Transformer-based encoder. The resulting hidden states are aggregated into a fixed-dimensional vector representation.

Dual-Encoder Structure TextCLIP adopts a dual-encoder architecture. The prompt encoder and response encoder are structurally similar but do not share parameters. Formally,

$$\mathbf{z}_p = \text{Encoder}_p(p), \quad \mathbf{z}_r = \text{Encoder}_r(r),$$

where $\mathbf{z}_p, \mathbf{z}_r \in \mathbb{R}^{d_0}$ are intermediate representations.

These intermediate representations are then projected into the shared embedding space through learnable projection heads:

$$\mathbf{h}_p = W_p \mathbf{z}_p + b_p, \quad \mathbf{h}_r = W_r \mathbf{z}_r + b_r,$$

where

$$W_p, W_r \in \mathbb{R}^{d \times d_0}, \quad b_p, b_r \in \mathbb{R}^d.$$

To stabilize training and make similarity scores comparable across examples, the projected vectors are normalized:

$$\tilde{\mathbf{h}}_p = \frac{\mathbf{h}_p}{\|\mathbf{h}_p\|_2}, \quad \tilde{\mathbf{h}}_r = \frac{\mathbf{h}_r}{\|\mathbf{h}_r\|_2}.$$

In the following, we use $\mathbf{h}_p$ and $\mathbf{h}_r$ to denote the normalized embeddings for simplicity.

3.4 SHARED EMBEDDING SPACE AND SIMILARITY FUNCTION

The learned embedding space is designed so that geometric proximity corresponds to semantic alignment. Given a prompt embedding $\mathbf{h}_{p_i}$ and a response embedding $\mathbf{h}_{r_{kj}}$, we define their similarity as scaled cosine similarity:

$$s(p_i, r_{kj}) = \frac{\mathbf{h}_{p_i}^\top \mathbf{h}_{r_{kj}}}{\tau},$$

where $\tau > 0$ is a temperature parameter.

This similarity is intended to measure how strongly a response reflects the prompt’s meaning and intention. It is therefore different from a scalar reward score used in standard reward modeling. Rather than assigning a global quality score to a response, TextCLIP evaluates the response relative to a specific prompt.

3.5 CONTRASTIVE LEARNING OBJECTIVE

Positive and Negative Pairs For each prompt p_i , the associated responses

$$\{r_{i1}, r_{i2}, \dots, r_{iK}\}$$

are treated as *positive* examples for that prompt. Responses generated for other prompts,

$$\{r_{kj} : k \neq i, j = 1, \dots, K\},$$

serve as *negative* examples.

This construction is central to our method. Because the positive responses come from multiple generators, they may differ substantially in wording, style, or length. What they share is that they are all responses to the same prompt. By contrast, the negative examples may be fluent and well-formed, but they correspond to different prompts. As a result, the model is encouraged to focus on prompt-specific semantic compatibility rather than on superficial properties that are not consistently tied to the prompt.

InfoNCE Loss For each prompt p_i and each matched response r_{ij} , we optimize an InfoNCE-style contrastive loss:

$$\mathcal{L}_{i,j} = -\log \frac{\exp(s(p_i, r_{ij}))}{\sum_{k=1}^N \sum_{m=1}^K \exp(s(p_i, r_{km}))}.$$

The full training objective is the average over all matched pairs:

$$\mathcal{L}_{\text{TextCLIP}} = \frac{1}{NK} \sum_{i=1}^N \sum_{j=1}^K \mathcal{L}_{i,j}.$$

Equivalently,

$$\mathcal{L}_{\text{TextCLIP}} = -\frac{1}{NK} \sum_{i=1}^N \sum_{j=1}^K \log \frac{\exp(s(p_i, r_{ij}))}{\sum_{k=1}^N \sum_{m=1}^K \exp(s(p_i, r_{km}))}.$$

This objective increases the similarity between each prompt and its matched responses while decreasing their relative similarity to responses from other prompts.

Multi-positive TextCLIP Loss. TextCLIP is trained with a multi-positive symmetric InfoNCE-style contrastive objective. Suppose a minibatch contains N prompts, and each prompt p_i is associated with a set of matched responses

$$\mathcal{R}_i^+ = \{r_{i1}, r_{i2}, \dots, r_{iK}\}.$$

Let

$$\mathcal{R} = \bigcup_{i=1}^N \mathcal{R}_i^+$$

denote the set of all responses in the minibatch. Given prompt and response embeddings z_{p_i} and z_r , we define the similarity score as

$$s(p_i, r) = \alpha z_{p_i}^\top z_r,$$

where α is a learnable temperature parameter.

For the prompt-to-response direction, all responses associated with the same prompt are treated as positives. The loss for prompt p_i is

$$\mathcal{L}_i^{p \rightarrow r} = -\log \frac{\sum_{r \in \mathcal{R}_i^+} \exp(s(p_i, r))}{\sum_{r \in \mathcal{R}} \exp(s(p_i, r))}.$$

The full prompt-to-response loss is

$$\mathcal{L}_{p \rightarrow r} = \frac{1}{N} \sum_{i=1}^N \mathcal{L}_i^{p \rightarrow r}.$$

For the response-to-prompt direction, each response r_{ij} is treated as positive for its corresponding prompt p_i and negative for other prompts in the minibatch. The loss for response r_{ij} is

$$\mathcal{L}_{ij}^{r \rightarrow p} = -\log \frac{\exp(s(p_i, r_{ij}))}{\sum_{k=1}^N \exp(s(p_k, r_{ij}))}.$$

The full response-to-prompt loss is

$$\mathcal{L}_{r \rightarrow p} = \frac{1}{\sum_{i=1}^N K} \sum_{i=1}^N \sum_{j=1}^K \mathcal{L}_{ij}^{r \rightarrow p}.$$

The final symmetric contrastive objective is

$$\mathcal{L}_{\text{TextCLIP}} = \frac{1}{2} (\mathcal{L}_{p \rightarrow r} + \mathcal{L}_{r \rightarrow p}).$$

This objective pulls each prompt toward all of its matched responses while pushing it away from responses associated with other prompts. At the same time, the symmetric response-to-prompt term encourages each response to be closer to its matched prompt than to other prompts in the minibatch. When each prompt has exactly one matched response, this objective reduces to the standard symmetric CLIP-style contrastive loss used in the single-positive setting.

Optimization The model parameters θ and ϕ are learned by minimizing

$$\min_{\theta, \phi} \mathcal{L}_{\text{TextCLIP}}.$$

Optimization is performed using stochastic gradient descent or Adam-style optimizers. The temperature parameter τ may be fixed or learned jointly with the encoder parameters.

4 EXPERIMENTS

4.1 DATASETS

We use two types of data in our experiments: (1) prompt–response data for contrastive pretraining, and (2) prompt–response–comparison data for pairwise evaluation and DPO fine-tuning.

Prompt–response pretraining data. We build a hybrid 20K prompt–response pairs from two SFT corpora: Smoltalk (allal et al., 2025) for reasoning and code tasks and AlpacaFarm (Dubois et al., 2023) for daily dialogues, and augment each prompt with three completions from DeepSeek-V3.1-Base (DeepSeek-AI et al., 2025), LLaMA3-72B (Grattafiori et al., 2024) and Qwen3-235B-A22B (Yang et al., 2025). Thus, each prompt has four responses, balancing semantic overlap and stylistic diversity to support learning invariant causal patterns. That is, if a prompt p_i is associated with responses $\{r_{i1}, r_{i2}, r_{i3}, r_{i4}\}$, then we construct multiple positive training pairs

$$(p_i, r_{i1}), (p_i, r_{i2}), (p_i, r_{i3}), (p_i, r_{i4}).$$

We also experiment with a multi-positive setting in which all responses associated with the same prompt are treated as positives simultaneously. This training structure is directly reflected in the data loading pipeline, where a prompt is paired either with one flattened response or with a list of multiple responses from the same prompt.

Pairwise comparison data. To evaluate the model’s ability to capture the on-topicalness of a prompt–response pair, we construct a subset of 300 preference pairs sampled from RewardBench2. For each chosen response, we construct three rewrites designed to isolate specific confounding factors:

- **Rewrite 1 (Lengthened):** We apply a RATE-style rewriting prompt to make the chosen response significantly longer, while preserving its factual content, stance, and topicality (?).
- **Rewrite 2 (Shortened):** Starting from Rewrite 1, we apply another RATE-style prompt to reduce its length, again without altering the original intent or content.
- **Rewrite 3 (Lengthened, Off-topic):** We generate a longer version of the chosen response that includes slight topical drift—maintaining politeness and fluency, but deviating from the core question or user intent.

For preference-style evaluation, each example has the form

$$(p_i, r_i^+, r_i^-),$$

where r_i^+ is the chosen response and r_i^- is the rejected response. For topic-sensitivity evaluation, we use a local dataset in which each prompt is paired with a chosen response and several alternative rewrites. From this dataset, we construct different test modes:

- **corr_rewrite:** the negative is a corresponding rewrite of the same prompt, designed to be a harder, more topic-related distractor;
- **random_rewrite:** the negative is a rewrite sampled from another example;
- **random_chosen:** the negative is a chosen response sampled from another prompt;
- **length_bias_rate:** both positive and negative are rewrite variants, enabling a stricter test of ranking behavior under controlled surface variation.

These splits are implemented directly in the dataset construction code through different sampling modes.

Additional evaluation sets. We also evaluate the model on two auxiliary pairwise alignment datasets. For the toxicity-related task, we use Toxic-DPO v0.2 (unalignment, 2024), a Hugging Face DPO-format dataset containing toxicity-related chosen–rejected response pairs. For the general human preference task, we use a binarized prompt–chosen–rejected version of UltraFeedback (Cui et al., 2024), a large-scale AI feedback dataset widely used for reward modeling and alignment. These datasets allow us to test whether the learned prompt–response compatibility score generalizes beyond on-topic evaluation to broader alignment settings.

4.2 MODEL

We evaluate two TextCLIP variants: an original lightweight model trained from scratch and a pretrained-encoder TextCLIP model initialized from a stronger semantic backbone. Both models follow the same dual-encoder principle: prompts and responses are encoded separately, projected into a shared latent space, normalized, and then compared using a scaled dot-product similarity.

Original TextCLIP. The original TextCLIP model is a lightweight dual-encoder contrastive model trained from scratch. Prompts and responses share a token embedding layer and a positional embedding, but they are passed through separate Transformer encoder stacks for prompt encoding and response encoding. This design preserves the asymmetric roles of prompts and responses while mapping both sides into a shared latent space.

In our implementation, the maximum context length is set to 1024, the batch size is 32, and the vocabulary size is 512. The token embedding dimension and Transformer hidden width are both 256. Each prompt and response encoder contains 4 Transformer layers with 8 self-attention heads. Each Transformer block consists of multi-head self-attention, layer normalization, and a feed-forward MLP whose hidden dimension is four times the Transformer width. The activation function in the feed-forward layer is QuickGELU. After the Transformer encoder, we apply mean pooling over the sequence dimension and use a linear projection head to map the pooled representation into a 256-dimensional latent space. The resulting prompt and response embeddings are normalized before similarity computation.

Formally, for a tokenized input $x = (x_1, \dots, x_T)$, the model first computes token and positional embeddings, applies the corresponding Transformer encoder, and then mean-pools the contextualized token representations:

$$\bar{h}(x) = \frac{1}{T} \sum_{t=1}^T h_t.$$

For prompt p and response r , the final normalized embeddings are

$$z_p = \frac{g_p(\bar{h}_p)}{\|g_p(\bar{h}_p)\|_2}, \quad z_r = \frac{g_r(\bar{h}_r)}{\|g_r(\bar{h}_r)\|_2}.$$

The compatibility score is computed as

$$s(p, r) = \alpha z_p^\top z_r,$$

where $\alpha = \exp(\gamma)$ is a learnable logit scale. In the original implementation, the logit-scale parameter is initialized to 0.07 before exponentiation. The token embedding weights are initialized from a normal distribution with standard deviation 0.02, the positional embedding is initialized with standard deviation 0.01, and the prompt and response projection weights are initialized with standard deviation $256^{-1/2}$. The model is trained using the symmetric contrastive objective described in the methodology section.

Pretrained TextCLIP. In order to show that the TextCLIP framework could be generalized to pretrained embedding models, we also conduct experiments on pretrained TextCLIP. The pretrained TextCLIP variant replaces the lightweight Transformer encoders with the pretrained model BAAI/bge-base-en-v1.5. Instead of training token embeddings and Transformer layers from scratch, this model uses the pretrained encoder to produce contextualized token-level representations for both prompts and responses. The pretrained encoder is shared across the two branches, while the prompt and response sides use separate projection heads. This allows the model to retain the

semantic knowledge of the pretrained encoder while adapting the final shared embedding space to prompt–response compatibility.

Given an input sequence, the pretrained encoder returns a sequence of hidden states. We aggregate these token-level representations using attention-mask-aware mean pooling:

$$\bar{h}(x) = \frac{1}{\sum_{t=1}^T m_t} \sum_{t=1}^T m_t h_t,$$

where m_t denotes the attention mask. This ensures that padding tokens do not contribute to the pooled representation.

On top of the pretrained encoder, we use two separate two-layer MLP projectors, one for prompts and one for responses. Each projector maps the encoder hidden size to a 512-dimensional latent space through a hidden layer of dimension 1024, GELU activation, and dropout rate 0.1:

$$g(\bar{h}) = W_2 \text{GELU}(W_1 \bar{h} + b_1) + b_2.$$

The final normalized embeddings are

$$z_p = \frac{g_p(\bar{h}(p))}{\|g_p(\bar{h}(p))\|_2}, \quad z_r = \frac{g_r(\bar{h}(r))}{\|g_r(\bar{h}(r))\|_2}.$$

The compatibility score is again computed as a scaled dot product:

$$s(p, r) = \alpha z_p^\top z_r.$$

For this pretrained variant, the learnable logit scale is initialized as

$$\gamma = \log(1/0.07), \quad \alpha = \exp(\gamma),$$

and is clamped during training so that $\alpha \leq 100$. By default, the pretrained encoder is fine-tuned jointly with the prompt and response projectors rather than frozen. The model supports both the standard symmetric contrastive loss and the multi-positive contrastive loss described in the methodology section. For pairwise preference experiments, we additionally compute chosen and rejected response scores using the dot product between the prompt embedding and each response embedding, and optimize a pairwise ranking loss when fine-tuning on chosen–rejected data.

4.3 TRAINING

The methodology section defines the multi-positive symmetric InfoNCE objective used to train TextCLIP. In this section, we describe how this objective is instantiated in our experiments, including the training data, minibatch construction, optimization settings, checkpoint selection, and early stopping criteria.

Training of Original TextCLIP. For Original TextCLIP, we train the lightweight dual-encoder model from scratch on `merged_dataset_20K.jsonl`. Inputs are tokenized with the dummy tokenizer and padded or truncated to a maximum context length of 1024. The batch size is set to 32, and the model is optimized with AdamW using a learning rate of 1×10^{-4} .

We consider both single-positive and multi-positive training settings. In the single-positive setting, each minibatch contains one matched response for each prompt, so the multi-positive objective reduces to the standard symmetric CLIP-style contrastive loss. In the multi-positive setting, each prompt is paired with multiple matched responses, and all responses associated with the same prompt are treated as positives. For this setting, the response tensor has shape $B \times K \times T$, where B is the batch size, K is the number of matched responses per prompt, and T is the sequence length.

For model selection, we save the checkpoint with the best training or validation objective and additionally monitor performance on the on-topic evaluation set. For the main multi-positive training run, we split the dataset into 90% training and 10% validation using a fixed random seed of 42. We evaluate validation loss and Recall@1, where a prediction is counted as correct if the top-ranked response belongs to the same prompt group. Training is run for up to 100 epochs with early stopping. We stop training when validation performance no longer improves for 5 epochs or when the normalized validation loss falls below a predefined threshold. To stabilize training, the learnable logit-scale parameter is frozen after several epochs.

Training of Pretrained TextCLIP. For Pretrained TextCLIP, we initialize the encoder from BAAI/bge-base-en-v1.5 and train it together with separate prompt and response projection heads. The input texts are tokenized with the pretrained model tokenizer, encoded by the BGE backbone, and then mapped into the shared TextCLIP space through the two MLP projectors.

We use the same contrastive training framework as in Original TextCLIP. In the single-positive setting, each prompt is paired with one matched response. When multiple matched responses are available, we use the multi-positive version of the objective, where all responses belonging to the same prompt are treated as positives. The response-to-prompt direction is implemented by flattening all responses in the batch and predicting the correct prompt index for each response. The learnable logit scale is initialized as $\log(1/0.07)$, exponentiated during training, and clamped so that the effective scale does not exceed 100.

For pairwise preference datasets with chosen and rejected responses, the pretrained model can additionally be fine-tuned with a pairwise ranking objective that encourages the chosen response to receive a higher prompt-compatibility score than the rejected response. Unless otherwise specified, however, the main TextCLIP results are trained with the contrastive objective so that Original TextCLIP and Pretrained TextCLIP are evaluated under the same training framework.

4.4 EVALUATION PROTOCOL

At test time, we independently encode the prompt, the chosen response, and the reject response, and compute cosine similarities

$$s^+ = z_p^\top z_{r^+}, \quad s^- = z_p^\top z_{r^-}.$$

A prediction is counted as correct if

$$s^+ > s^-.$$

We report pairwise accuracy as the main metric. For each evaluation setting, we also save the full distributions of chosen and reject similarities and visualize them with scatter plots of chosen versus reject scores. This evaluation procedure is shared across the preference, on-topic, toxicity, and length-bias settings.

4.5 RESULTS

Training Results The training loss of original TextCLIP according to epoch is shown as Fig 2, which keeps decreasing and converges near 0.2 at the end. The three on-topic validation accuracy across epochs of the original TextCLIP is shown as Fig 3. They all have an increasing trend and achieve a good performance in the end.

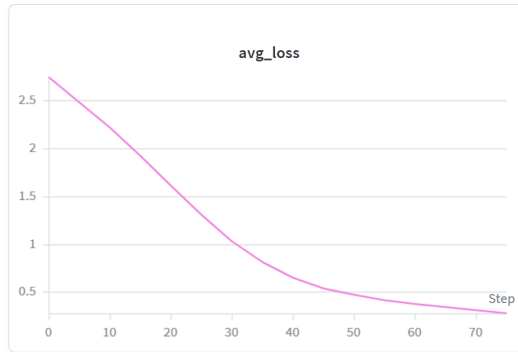


Figure 2: Training loss of original TextCLIP.

Table 1 shows that pretrained TextCLIP achieves the best performance across all three on-topic evaluation settings. The largest gap appears on the hardest setting, *corr-rewrite*, where Original TextCLIP, pretrained embedding similarity, and pretrained TextCLIP achieve accuracies of 0.768, 0.812, and 0.856, respectively. This suggests that pretrained semantic representations already provide a strong signal, while contrastive learning can further adapt the representation space for prompt-response alignment.

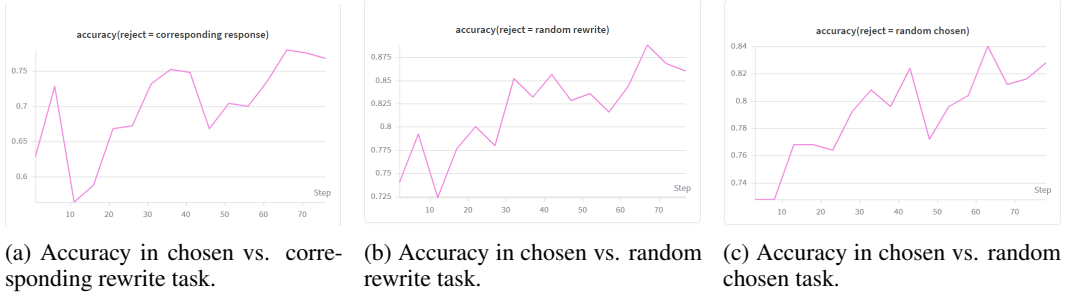


Figure 3: Validation accuracy across training epochs.

On the easier *random-rewrite* and *random-chosen* settings, all methods perform strongly, with the pretrained methods reaching near-perfect accuracy. This indicates that these settings are dominated more by coarse semantic relatedness, whereas *corr-rewrite* provides a more challenging and informative evaluation of fine-grained alignment.

Table 1: Accuracy on the ontopic corresponding rewrite task.

Method	corr-rewrite	random-rewrite	random-chosen
Original TextCLIP	0.768	0.856	0.917
Pretrained Embedding Similarity	0.812	1.000	0.988
Pretrained TextCLIP	0.856	1.000	0.993

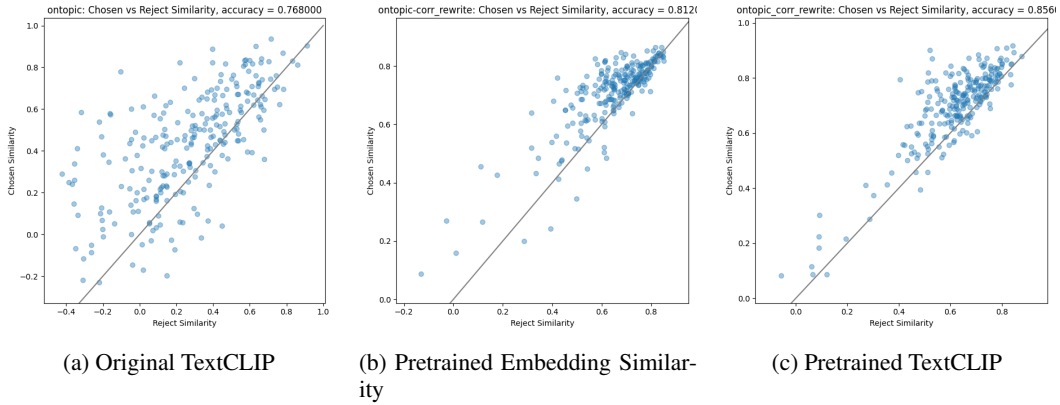


Figure 4: Similarity plots for the ontopic corresponding rewrite task.

Interpretation of the results. The original TextCLIP uses a relatively small task-specific encoder. Its embedding layer is an `nn.Embedding(vocab_size, transformer_width)`, and under our current setting this corresponds to a token embedding matrix of size 256×512 . On top of this embedding layer, the model only uses lightweight prompt and response Transformers, followed by linear projection heads that map representations into a 512-dimensional latent space. In contrast, the pretrained variants start from BAAI/bge-base-en-v1.5, whose contextual hidden size is 768, and then use a two-layer MLP projector of size $768 \rightarrow 1024 \rightarrow 512$.

Despite this substantial gap in encoder strength, the original TextCLIP still achieves an accuracy of 0.768 on the ontopic corresponding rewrite task, which is already a strong result for such a simple and parameter-efficient model. This suggests that the contrastive learning framework itself is able to learn a meaningful prompt-response alignment space. The pretrained embedding similarity baseline further improves the accuracy to 0.812, showing that pretrained semantic representations already provide a strong signal for this task. However, pretrained TextCLIP reaches the best result, 0.856,

indicating that contrastive learning remains useful even on top of a strong pretrained embedding model and can further adapt the embedding space for prompt–response alignment.

Table 2: Length-bias ATE evaluation under two longer-response settings.

Setting	ATE	95% CI
Shorter vs. longer rewrite	−0.019	[−0.040, 0.001]
Shorter vs. longer off-topic rewrite	−0.140	[−0.164, −0.113]

Table 3: Length-bias correlation evaluation under two longer-response settings.

Setting	Pooled Pearson	Pooled Spearman	Δ Pearson	Δ Spearman
Shorter vs. longer rewrite	0.149 ($p < 0.001$)	0.133 ($p = 0.003$)	−0.038 ($p = 0.547$)	−0.006 ($p = 0.924$)
Shorter vs. longer off-topic rewrite	−0.026 ($p = 0.561$)	−0.037 ($p = 0.411$)	−0.109 ($p = 0.086$)	−0.074 ($p = 0.244$)

Length-bias evaluation Table 2 and Table 3 compares two length-related evaluation settings.

In the first setting, we compare a shorter rewrite with a longer rewrite. Both rewrites maintain the content quality and topic as the original response. The estimated ATE is close to zero and not statistically significant ($\widehat{\text{ATE}} = -0.019$, 95% CI [−0.040, 0.001]), indicating that the longer rewrite does not receive a significantly higher compatibility score on average. Although the pooled length–score correlation is weakly positive and statistically significant (Pearson $r = 0.149$, $p < 0.001$), the prompt-controlled correlation between within-prompt length increase and score increase is not significant ($r = -0.038$, $p = 0.547$). This suggests that the aggregate correlation may reflect prompt- or content-level variation rather than a direct preference for longer responses.

In the second setting, we compare the shorter rewrite with a longer rewrite that introduces off-topic content. Here, the ATE is strongly negative ($\widehat{\text{ATE}} = -0.140$, 95% CI [−0.164, −0.113]), showing that the longer off-topic rewrite receives substantially lower compatibility scores on average. The pooled length–score correlation is not significant, and the prompt-controlled Δ length– Δ score correlation is negative but not significant at the 0.05 level. Together, these results suggest that TextCLIP does not simply reward longer responses. Instead, when additional length is accompanied by reduced prompt-specific relevance, the model assigns lower compatibility scores.

4.6 FINE-TUNING RESULTS

To further examine whether the TextCLIP framework can be adapted to different alignment-related objectives, we fine-tune the model on three pairwise preference tasks: on-topic response selection, toxicity-related preference, and general human preference. For each task, the data are organized as prompt–chosen–rejected triples. Starting from the contrastively trained TextCLIP model, we apply DPO-style fine-tuning to encourage the model to assign a higher compatibility score to the chosen response than to the rejected response.

We evaluate all models using pairwise accuracy. Given a prompt p , a chosen response r^+ , and a rejected response r^- , the model predicts correctly if $s(p, r^+) > s(p, r^-)$. This evaluation protocol is shared across all three fine-tuning tasks, allowing us to directly compare whether DPO fine-tuning improves the prompt–response compatibility score for different forms of alignment.

Table 4: Pairwise accuracy before and after DPO fine-tuning across three alignment tasks.

Task	Training Stage	DPO Stage
On-topic response selection	76.9%	94.8%
Toxicity-related preference	80.4%	99.6%
Human preference	61.7%	87.7%

Table 4 shows that DPO fine-tuning substantially improves pairwise accuracy across all three alignment tasks. For on-topic response selection, accuracy increases from 76.9% to 94.8%, indicating

that DPO fine-tuning strengthens the model’s ability to distinguish responses that better address the prompt intention. For toxicity-related preference, accuracy increases from 80.4% to 99.6%, suggesting that the TextCLIP scoring function can be strongly adapted to safety-related preference distinctions. For general human preference, accuracy improves from 61.7% to 87.7%, showing that the framework also generalizes beyond task-specific semantic relevance to broader preference judgments.

Overall, these results suggest that the shared prompt–response embedding space is not fixed after contrastive training, but can be further reshaped by preference optimization. DPO fine-tuning consistently increases the separation between chosen and rejected responses under the TextCLIP scoring function, supporting the view that TextCLIP can serve as a flexible representation-learning framework for different alignment objectives.

4.7 SHARED EMBEDDING SPACE ANALYSIS

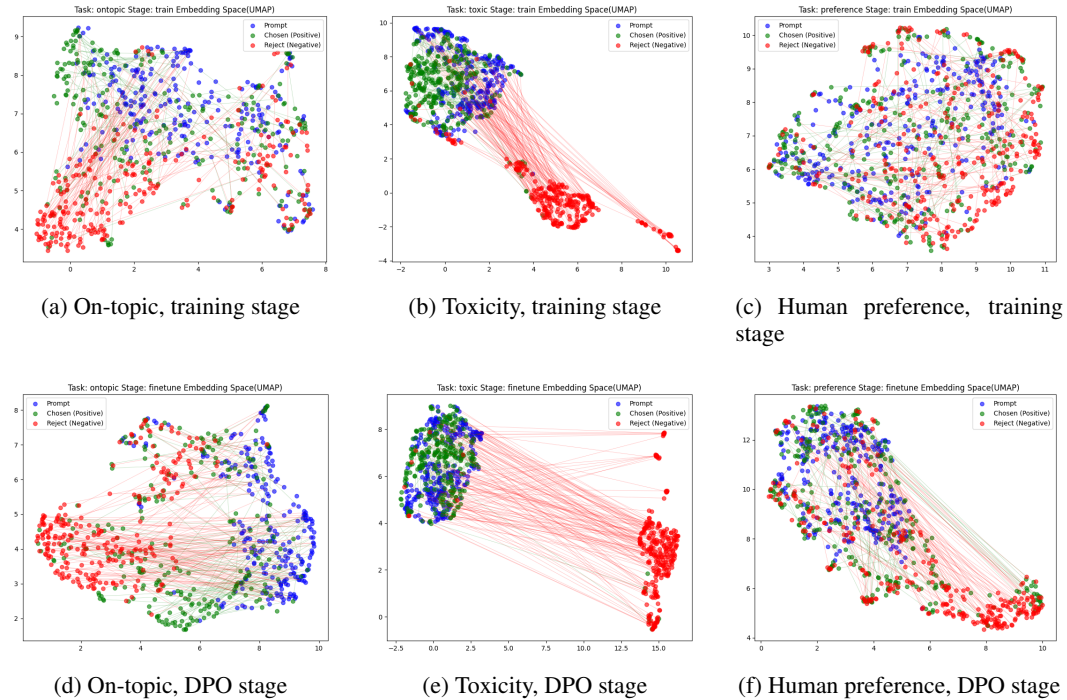


Figure 5: UMAP visualizations of the shared prompt–response embedding space across three alignment tasks before and after DPO fine-tuning. The first row shows the embedding space after contrastive training, and the second row shows the embedding space after DPO fine-tuning. Blue points denote prompts, green points denote chosen responses, and red points denote rejected responses.

Figure 5 visualizes the shared embedding space by applying UMAP to the learned prompt, chosen-response, and rejected-response embeddings for each task and plotting the first two UMAP components before and after DPO finetuning.

The first row shows the contrastive training stage, and the second row shows the DPO fine-tuning stage. Blue points denote prompts, green points denote chosen responses, and red points denote rejected responses.

In the training stage, TextCLIP already learns a meaningful alignment structure. For on-topic task, Figure 5b provides a qualitative visualization of the learned shared embedding space. Chosen responses are generally embedded closer to their corresponding prompts, while rejected responses are more often located farther away. The visualization also shows a clear separation between the chosen and rejected response regions: chosen responses tend to overlap with the prompt region, whereas rejected responses form a more distinct lower-left cluster. This suggests that TextCLIP learns an embedding geometry that reflects prompt–response compatibility, with preferred responses placed

closer to prompt representations and rejected responses pushed away. For the other 2 tasks, the plots also shows good performances. The toxicity-related task shows an even clearer separation between chosen and rejected responses, indicating that safety-related preference distinctions are strongly encoded in the embedding space. The human-preference task exhibits more overlap, which is expected because general preference is more heterogeneous and may depend on multiple factors such as relevance, helpfulness, style, and specificity.

After DPO fine-tuning, the separation between chosen and rejected responses becomes more pronounced across all three tasks. This is especially clear in the toxicity-related task, where rejected responses form a distinct cluster far from the prompt and chosen-response region. For the on-topic and human-preference tasks, DPO also sharpens the representation space by making chosen and rejected responses more distinguishable. Together with the accuracy improvements in Table 4, these visualizations suggest that DPO fine-tuning not only improves pairwise prediction accuracy, but also reshapes the shared embedding geometry by pulling chosen responses closer to prompts and pushing rejected responses farther away.

5 CONCLUSIONS

This thesis proposes TextCLIP, a contrastive representation learning framework for modeling prompt–response alignment. Rather than treating response quality as an absolute property, TextCLIP learns a prompt-conditioned compatibility function that measures whether a response addresses the intention, constraints, and semantic goal of a given prompt.

The experiments show that TextCLIP can learn a meaningful shared embedding space for prompts and responses. On the on-topic corresponding-rewrite task, the original lightweight TextCLIP achieves strong performance despite being trained from scratch, while the pretrained TextCLIP further improves accuracy and outperforms the pretrained embedding similarity baseline. These results suggest that contrastive learning can adapt both lightweight and pretrained representations toward prompt-specific alignment.

The length-bias analysis further indicates that TextCLIP does not simply reward longer responses. When longer rewrites preserve the original content, the average score difference is close to zero; when longer rewrites introduce off-topic content, they receive substantially lower compatibility scores. This supports the claim that TextCLIP is sensitive to semantic fit rather than only surface-level length.

Finally, DPO-style fine-tuning improves pairwise accuracy across on-topic, toxicity-related, and general preference tasks. The UMAP visualizations show that fine-tuning also reshapes the shared embedding space by making chosen and rejected responses more separable. Overall, these findings suggest that TextCLIP provides a useful and interpretable framework for studying prompt-specific alignment, with potential applications in response ranking, reward-model diagnostics, and future policy optimization.

A APPENDIX

A.1 REWRITING INSTRUCTION

We describe the rewriting strategy to construct the evaluation data sets for on-topic task in Table 5 followed by a specific example of the rewriting results A.1.

Rewriting Example

Prompt: *Why are the Kardashians so famous?*

Rewriting Type	Longer Rewrite	Shorter Rewrite	Longer Off-topic Rewrite
Rewriting Instruction	Rewrite the response to be longer while preserving meaning and style.	Make the previous response more concise without losing core information.	Expand the response with stylistic elaboration, allowing mild topical drift.

Table 5: Rewriting instructions used to construct response variants for robustness testing. Each instruction targets different attributes such as response length or topicality.

Longer Rewrite: The Kardashian family has undeniably emerged as a prominent and influential cultural phenomenon within contemporary American society. They are most notably represented by the reality television show *Keeping Up with the Kardashians*, which has garnered widespread critical acclaim and achieved record-breaking ratings throughout its various seasons and spin-offs. This show has not only captivated audiences but has also significantly shaped popular culture and social dynamics. Beyond their television presence, the Kardashian family has successfully ventured into various other fields, with multiple members pursuing thriving careers in fashion, entertainment, social media, and other diverse arenas. Their entrepreneurial spirit and ability to capitalize on their fame have led to the creation of numerous brands and products that resonate with fans and consumers alike. Overall, the Kardashian family serves as a striking representation of the increasing prevalence of celebrity and media culture in contemporary American society. Their influence underscores the importance of images and narratives in modern cultural discourse, illustrating how celebrity status can shape public perception and societal trends.

Shorter Rewrite: The Kardashian family has become a significant cultural phenomenon in contemporary American society, primarily through their reality show *Keeping Up with the Kardashians*, which has received critical acclaim and achieved record-breaking ratings. This show has not only captivated audiences but also shaped popular culture and social dynamics. Beyond television, the family has successfully ventured into various fields, with members building thriving careers in fashion, entertainment, and social media. Their entrepreneurial spirit has led to numerous brands and products that resonate with fans and consumers. Overall, the Kardashian family exemplifies the rise of celebrity and media culture in modern America, highlighting how celebrity status can influence public perception and societal trends.

Longer Off-topic Rewrite: The Kardashian family has undeniably emerged as a prominent and influential cultural phenomenon within contemporary American society, captivating audiences with their unique blend of glamour, drama, and entrepreneurial spirit. Most notably represented by the reality television show *Keeping Up with the Kardashians*, which has garnered widespread critical acclaim and achieved record-breaking ratings across its various seasons, the family’s impact extends far beyond the confines of the small screen. Each member of the family has carved out a niche for themselves, pursuing successful careers in diverse fields such as fashion, entertainment, and social media, thereby illustrating the multifaceted nature of their influence. This phenomenon raises intriguing questions about the evolving landscape of celebrity culture and the ways in which images and narratives shape public perception. Moreover, one might consider how the rise of social media platforms has transformed the way we engage with celebrities, blurring the lines between public and private life, and fostering a culture of constant connectivity. It’s fascinating to think about how this shift has not only affected the Kardashians but also countless other public figures who navigate the complexities of fame in the digital age. Overall, the Kardashian family’s prominence serves as a reflection of the increasing prevalence of celebrity and media culture in contemporary

American society, highlighting the significance of visual storytelling and the narratives we construct around public personas. What does this say about our collective values and the way we consume media?

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